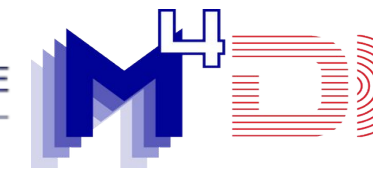




PROGRAMME
DE RECHERCHE
SANTÉ
NUMÉRIQUE



Introduction SNDS: Système National des Données de Santé

6 march 2025

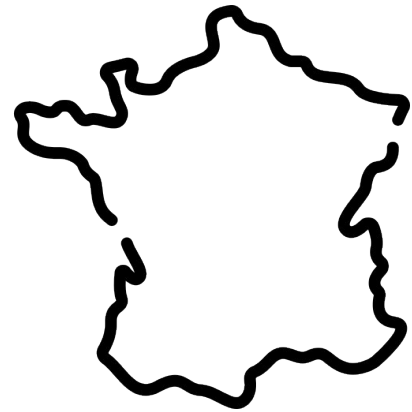
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Renée Le Clech

supervized by Anne-Sophie Jannot, Nicolas Garcelon, Nicola Jay and Aurélie Bannay

Systeme National des Données de Santé:



- > Pseudonymized medico-administrative data base.
- > Contains all treatments submitted for reimbursement for over 99% of the French population.
- > Managed by CNAM.



Some numbers about the SNDS:

- **1,2 billion** medical claim forms
- **11 million** hospital stays
- **500 million** medical procedures
- **450 TB** of data

(source: Health Data Hub)

Coming soon :

Medico-social data
Disability data
Some data from complementary
health insurance

2018: Death causes

2016 Creation of the SNDS, *Law for the modernization of the health system*

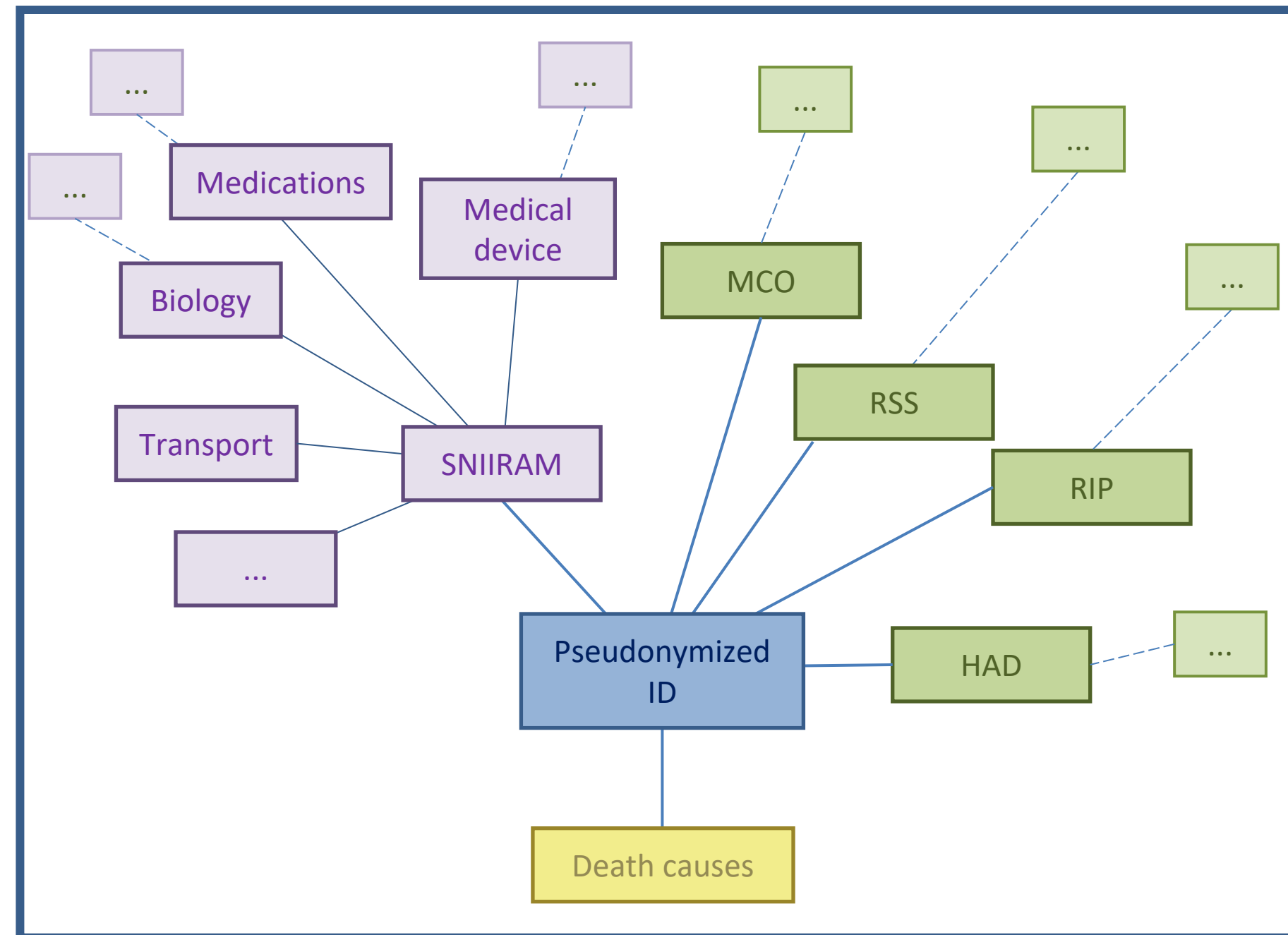
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Système national des données de santé



managed by the Caisse Nationale de l'Assurance Maladie (CNAM)

Link SNDS x other data bases
Access to the SNDS

Purposes of the SNDS

- Information on **health** and **healthcare provision**
- Evaluation of healthcare **policies**
- Evaluation of healthcare **expenditure**
- Information for healthcare **professionals** on their activities
- Health **surveillance** and **safety**
- Health **research, studies, evaluation** and **innovation**

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- **Promoting healthcare products** to healthcare professionals and institutions
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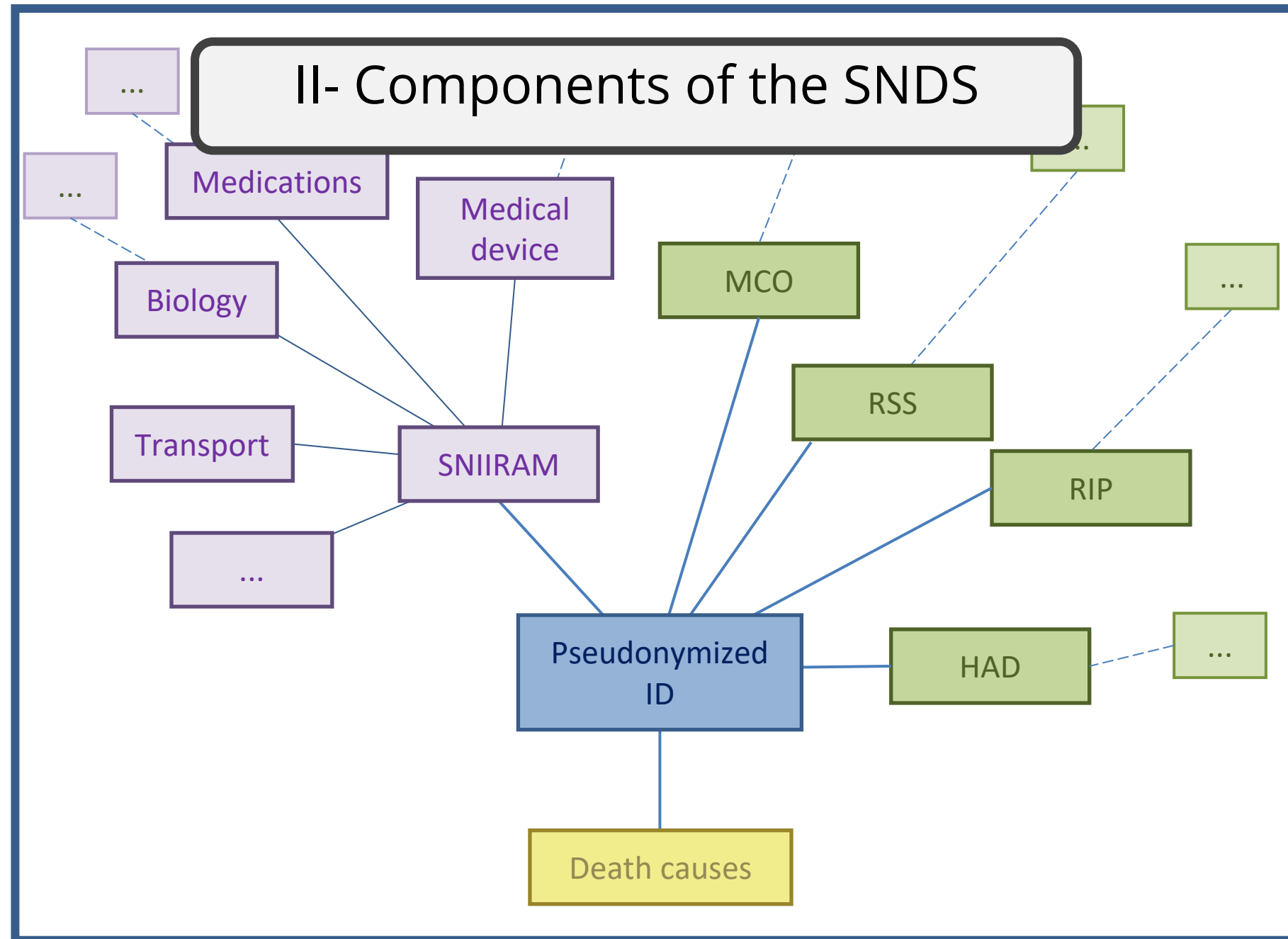
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Système national des données de santé

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A national spending target is set each year in the French Social Security Finance Act.

=> **need to monitor health insurance expenditure**

1996

Addition of some special schemes :

RSI- independent workers (5%),
MSA - farmers (5%),
Others (SNCF,RATP) (4%)

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Creation of the SNIIRAM: public health insurance data

Goal:

Improve **quality of care**

Contribute to better management of health **insurance** and **policies**

Provide healthcare **professionals** with relevant information on their activities

Initially concerns only the general scheme (~ 85% of the population)

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Hospital data added:
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2007 first linkages

2010 linkages completed

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Death dates

2009 For the general scheme
2013 For almost all others schemes

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Law for the modernization of the health system

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Death causes

Death causes implemented by the INSERM (CépiDC)

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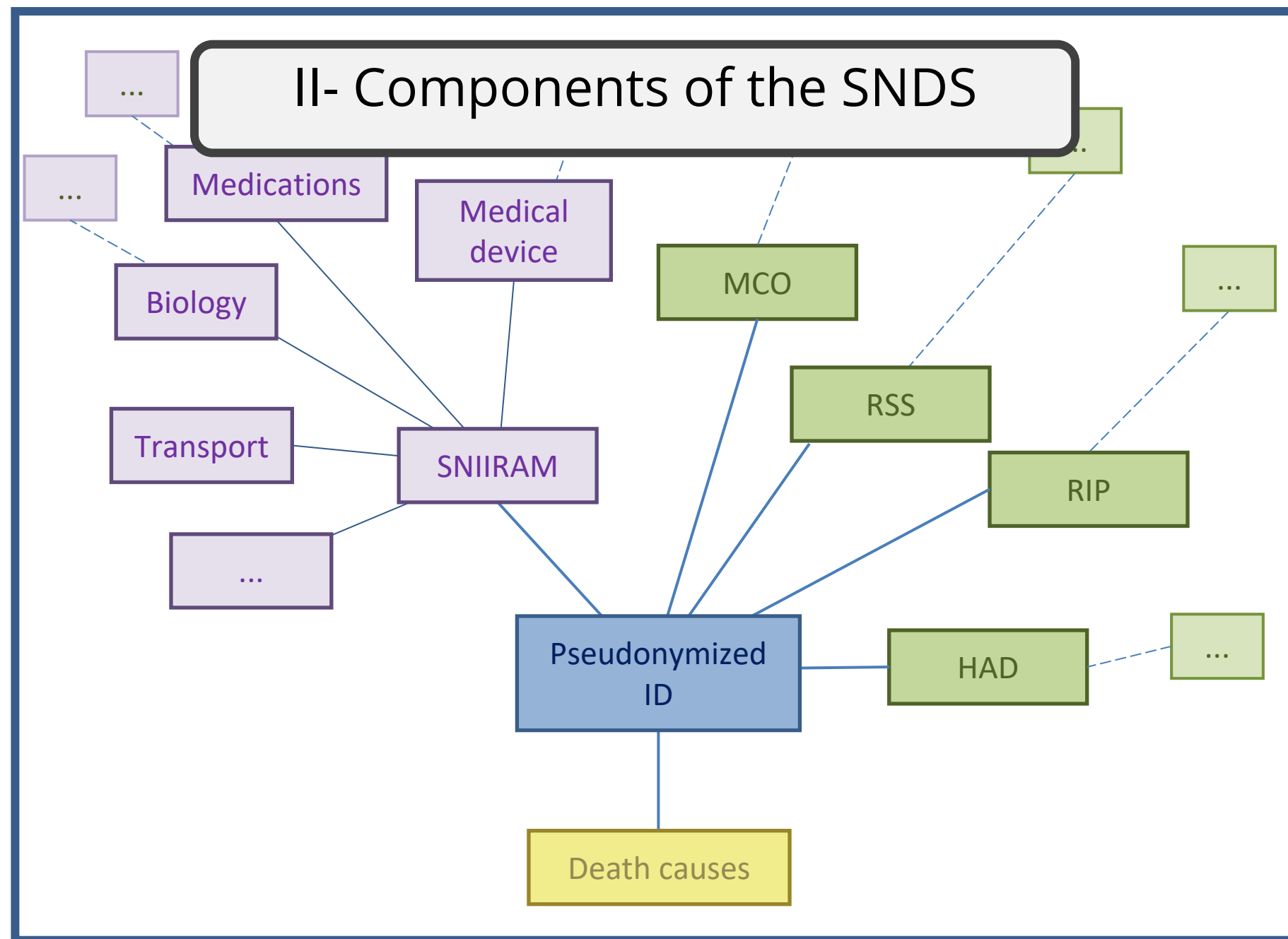
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Système national des données de santé

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managed by the Caisse Nationale de l'Assurance Maladie (CNAM)

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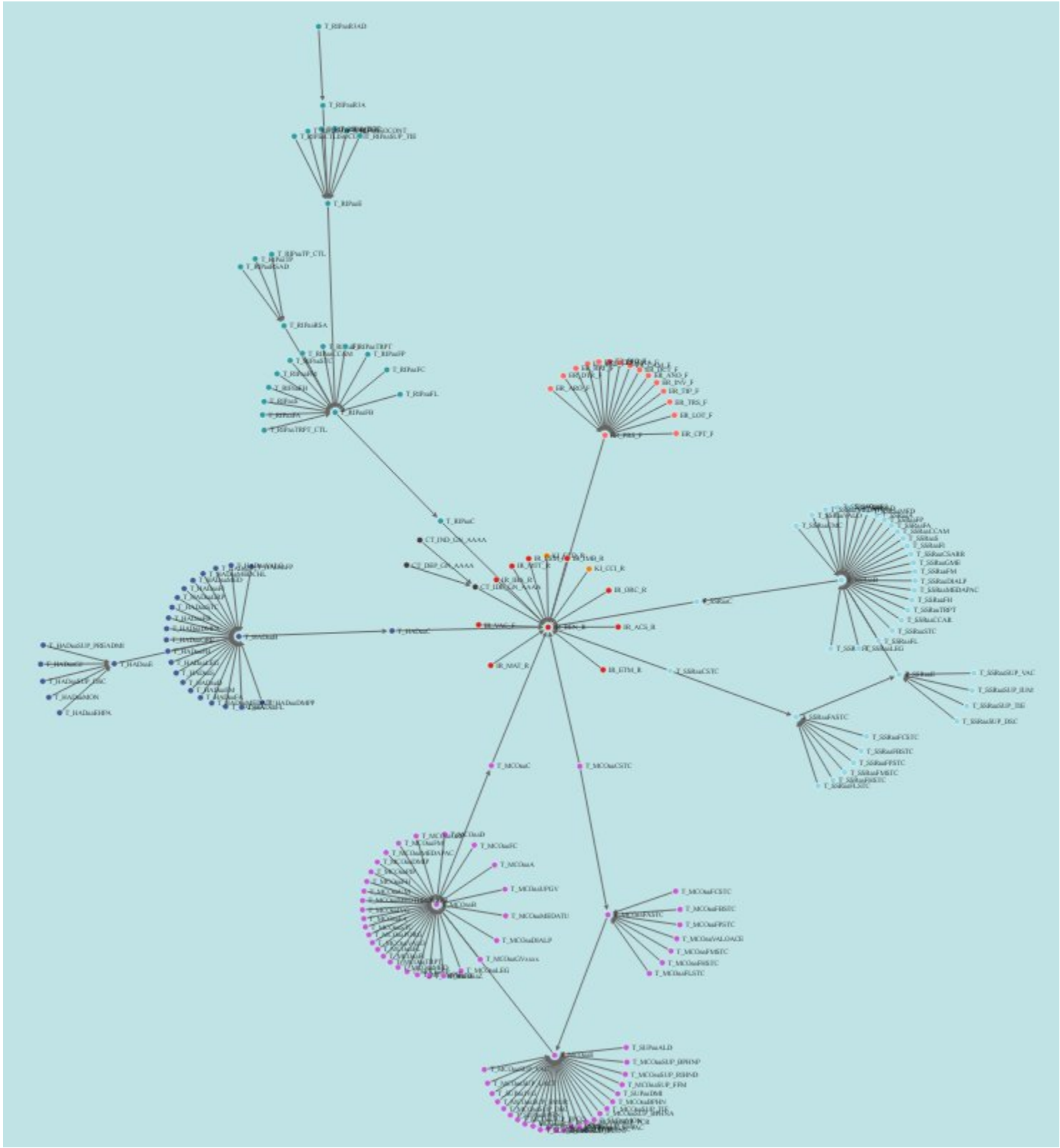
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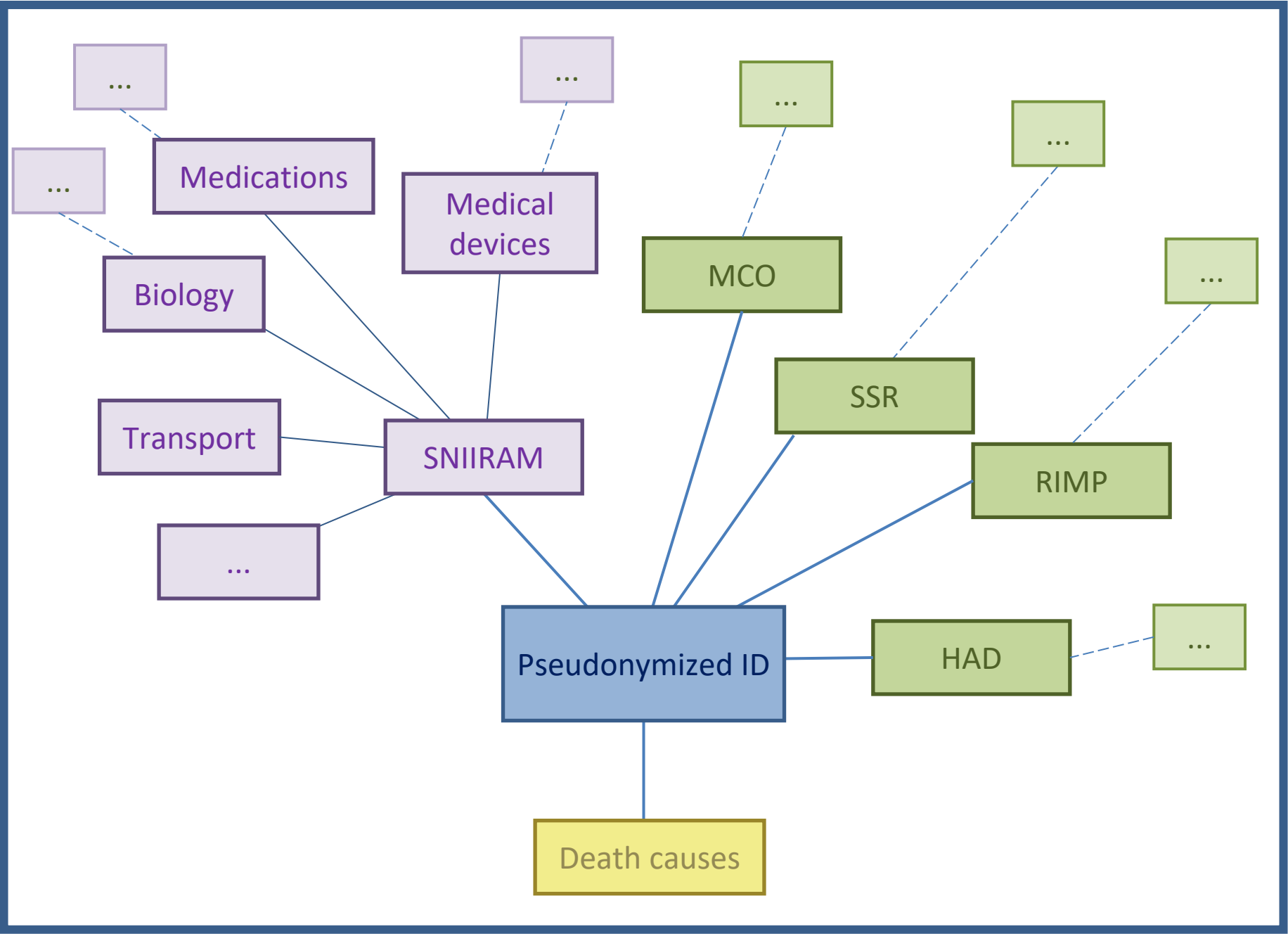
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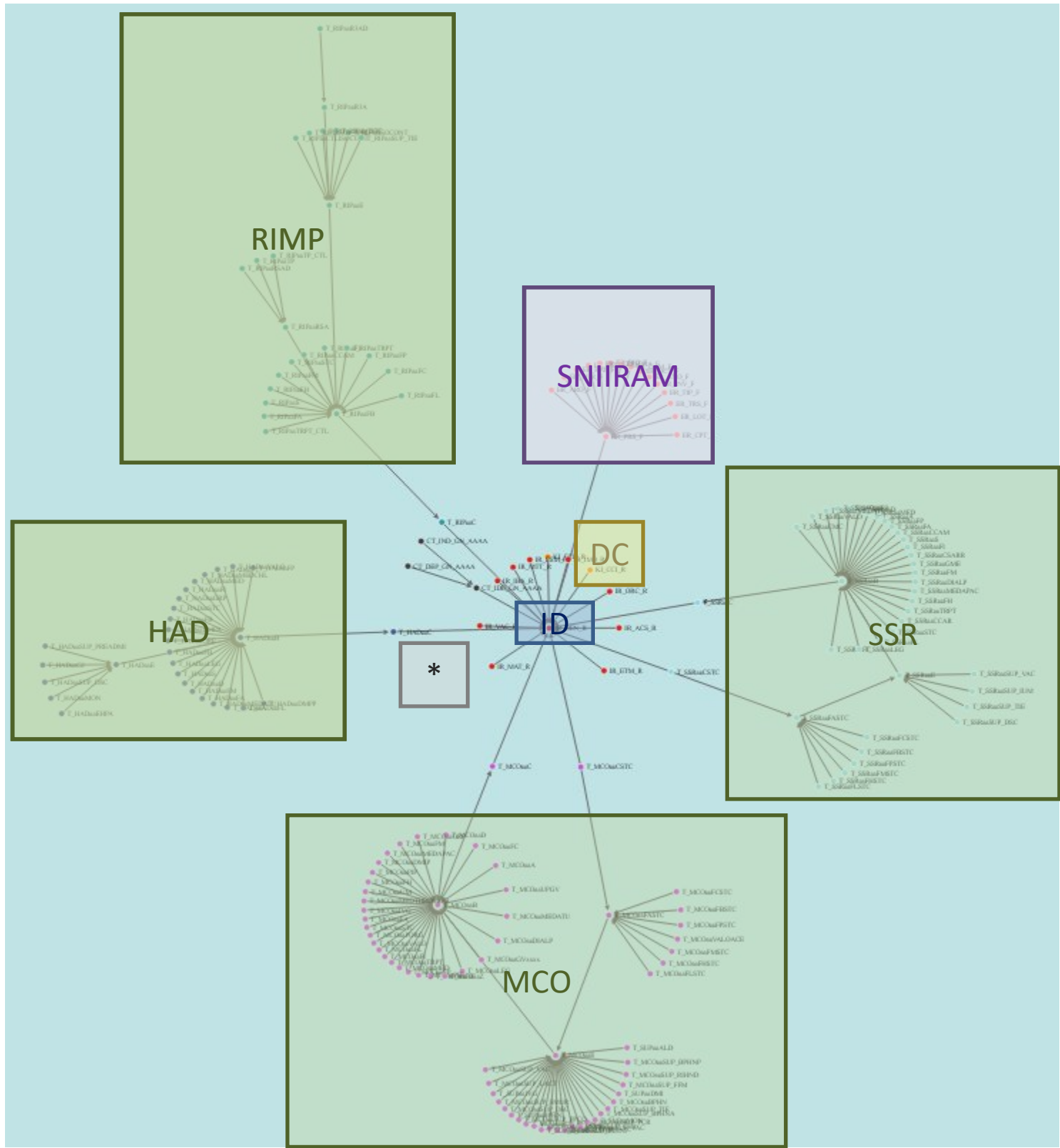


3 Screen shot of a interactive graph created by the *Health Data Hub*, representing the architecture of the SNDS data base

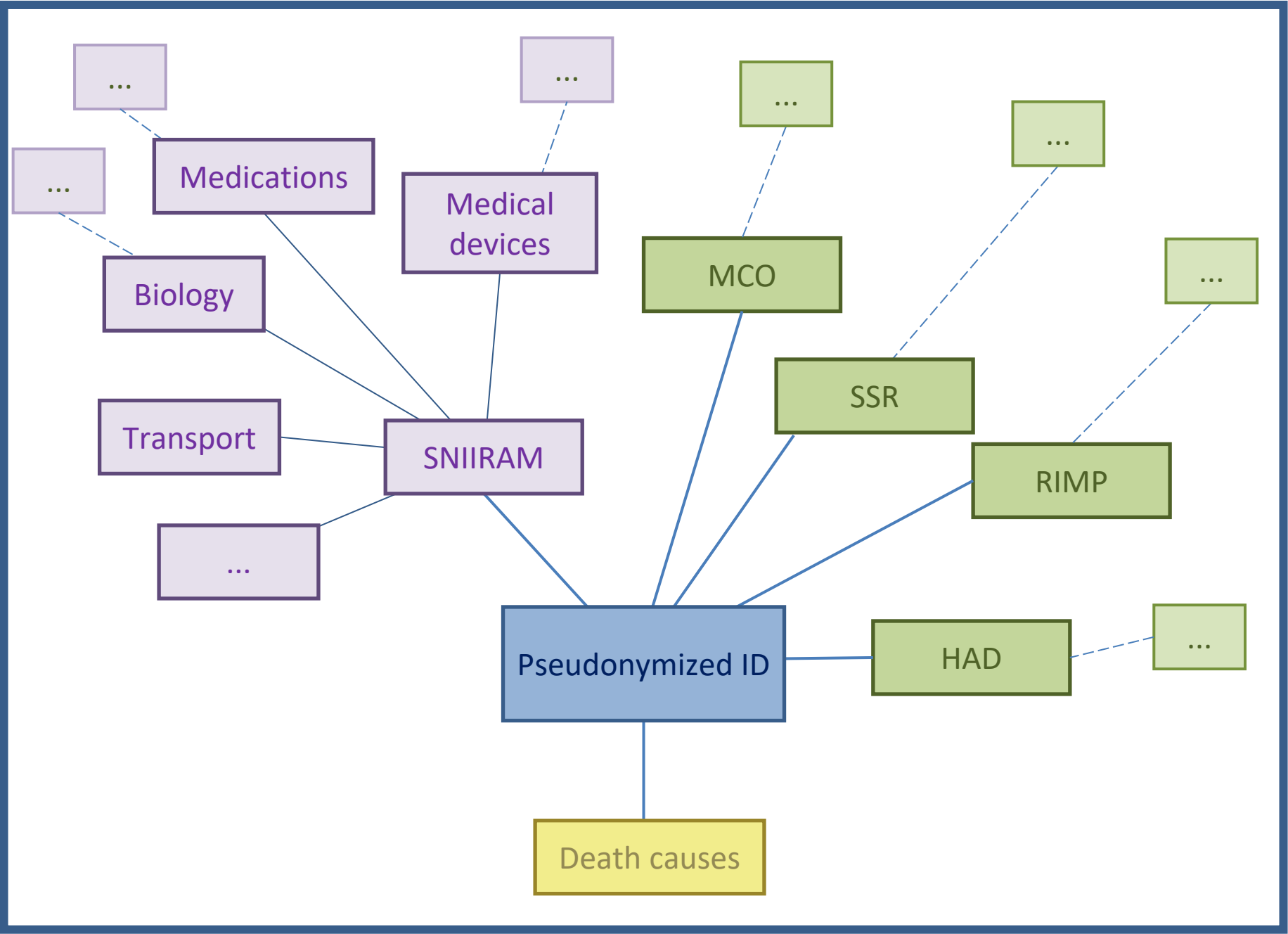


Simplified representation of the SNDS

II- Components of the SNDS



- * Other tables:
 - Table for COVID-19
 - Table of the referring doctors
 - Table for maternity data
 - Table of the affiliation to a complementary organization

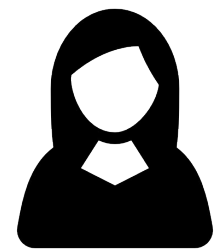


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Simplified representation of the SNDS

II- Components of the SNDS

Information we have :



Socio-demographic data

- Age, sex, place of living
- Income & Mean salary
- Long-term condition (CIM10)
- Assignment scheme (general, farmer , ...)



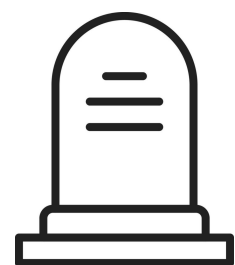
Non-hospital treatments data

- Name of the procedure/ product/ service/ medication/ consultation
- Reimbursement **date** and **amount**



Hospital data

- Principal & associated **diagnosis**
- Medical Unit
- Medical **acts** performed
- Date and place of the care
- **Expensive** medicines dispensed



Death causes data

- Death date
- Main cause and associated cause

IR BEN
Pseudonymized Id

SNIIRAM

MCO

SSR

RIMP

HAD

Death cause

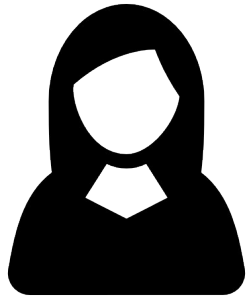


PMSI

Programme de
médicalisation des
systèmes d'information

II- Components of the SNDS

Information we don't have :



- Information on **employment/occupational activity**
- **Risk factors** (tobacco, alcohol, nutrition, etc.)



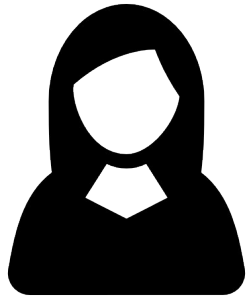
- Non-reimbursed drugs
- **Taking or not** taking medication (dispensing only)
- **Reason** for medical/paramedical consultation
- Test **results** (serology, CT scan/ X-ray images, etc.)



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- **Emergency** (beside hospitalization)
- **Palliative** care

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Data with **strengths**: almost an *exhaustive coverage* of the French population, *longitudinal data*
But also **weaknesses**: some data are *not present*, *complexity* of data, difficulty of *access*

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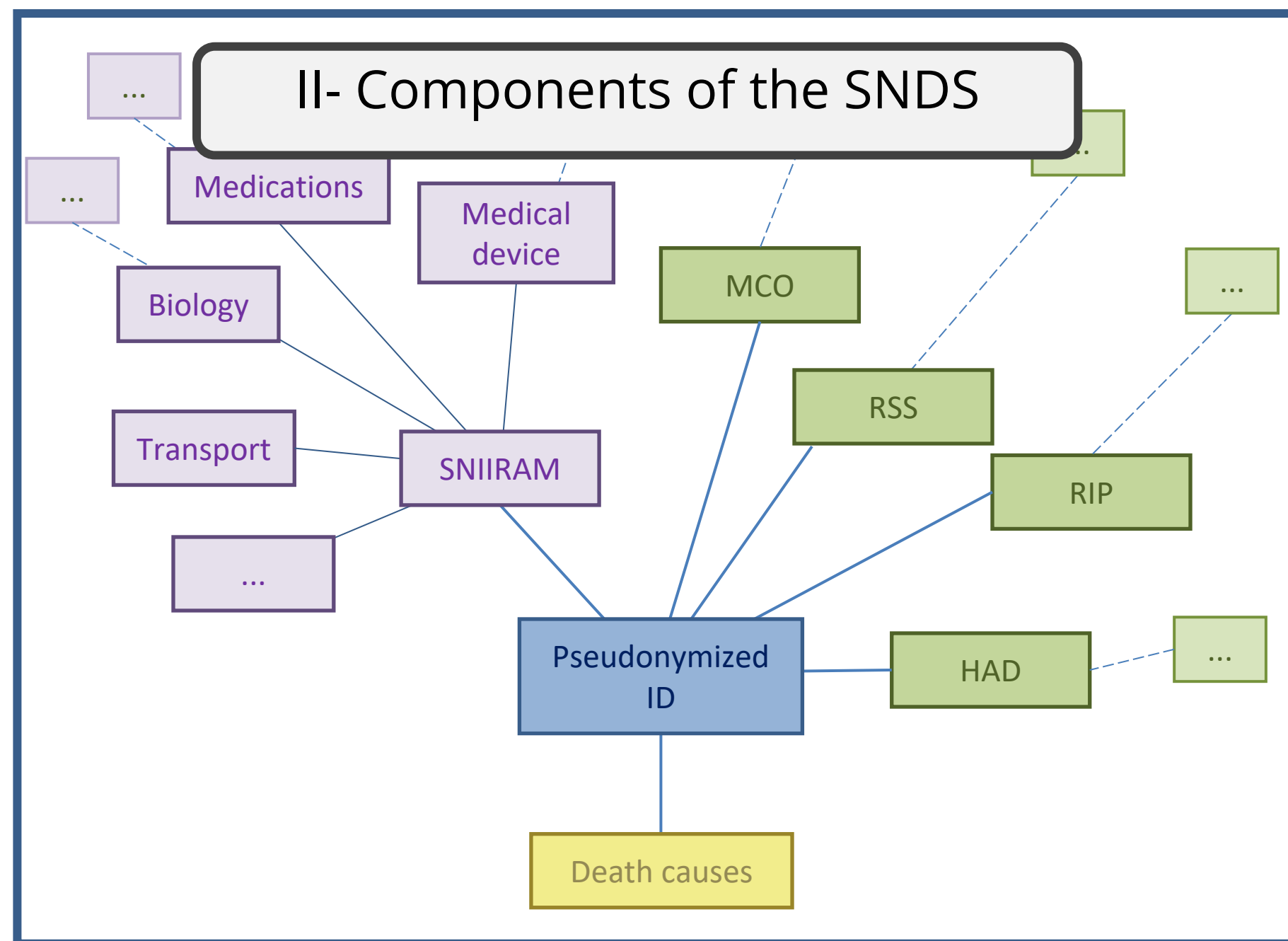
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Système national des données de santé

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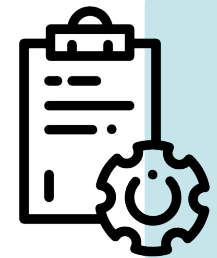
III- Access and linkage in the SNDS



Open data

All

Aggregated data, with no possibility of re-identification
Available on Ameli.fr, server open de l'AM, CépiDC,...



Projet access

After authorization from the CNIL
Commission Nationale de l'Informatique et des Libertés
(French Data Protection Authority)

Public or private player for a project of public interest.

> Access to the SNDS by healthcare manufacturers and insurers is more closely supervised. The CNIL has drawn up an ***exhaustive list of organizations*** committed to SNDS best practices.



Permanent access

Structures entrusted with public service missions

> List by decree (INSERM, INRIA, Institut National du Cancer, ARS, etc.).

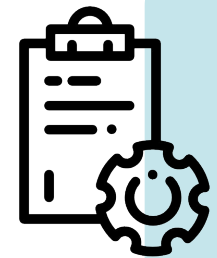
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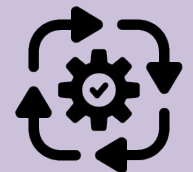
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(*terms and conditions*)

Mandatory
trainings

Project access :
Long procedure
(months or even years)



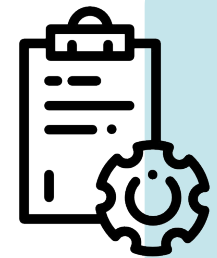
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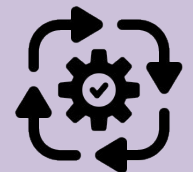
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Strong **traceability** of events relating to system security.

Possible a posteriori **sanctions** (fines) for non-compliance with T&Cs: re-identification of a patient

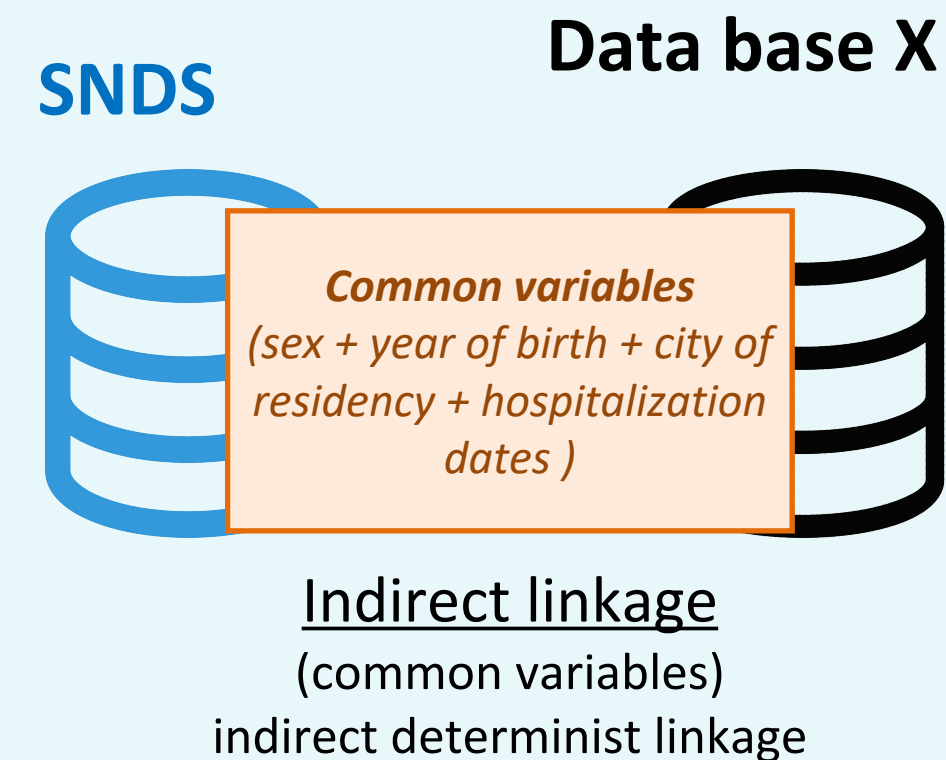
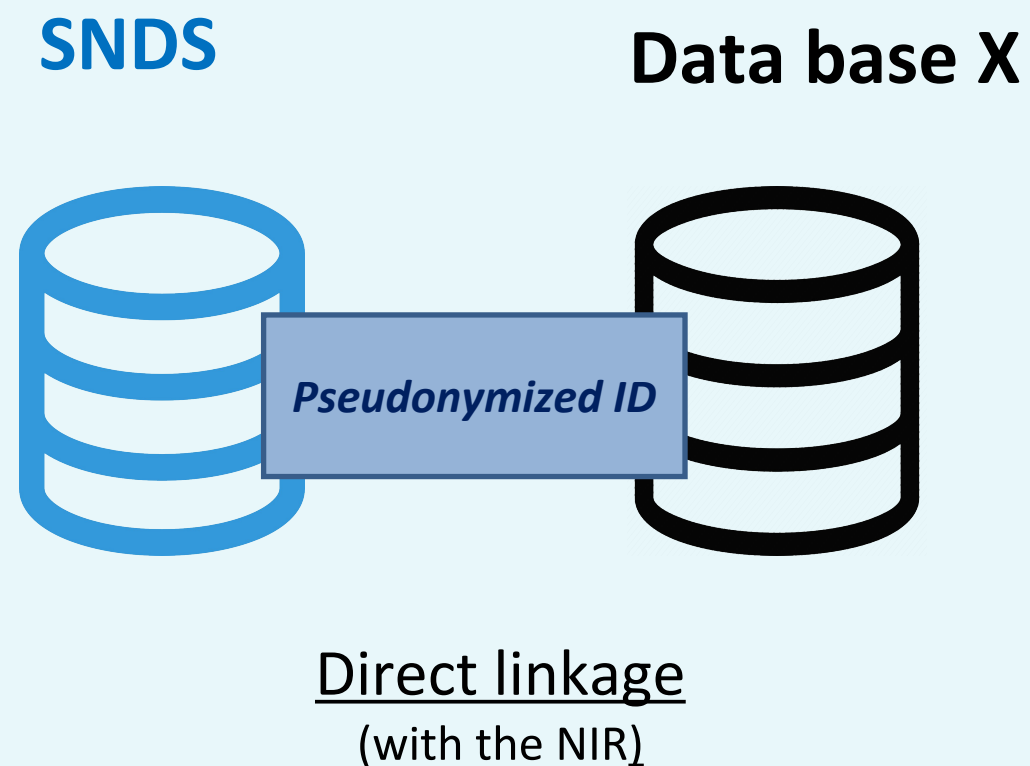
5 particularly sensitive variables:

Municipality of residence, Month and year of birth, Date of care, Date of death, Municipality of death.

III- Access and linkage in the SNDS

Linkage between data bases and the SNDS

Advantages: Enables existing databases to be supplemented with data on healthcare consumption from the SNDS. (ex : linkage between SNDS and rare diseases data base - Dylan and me)



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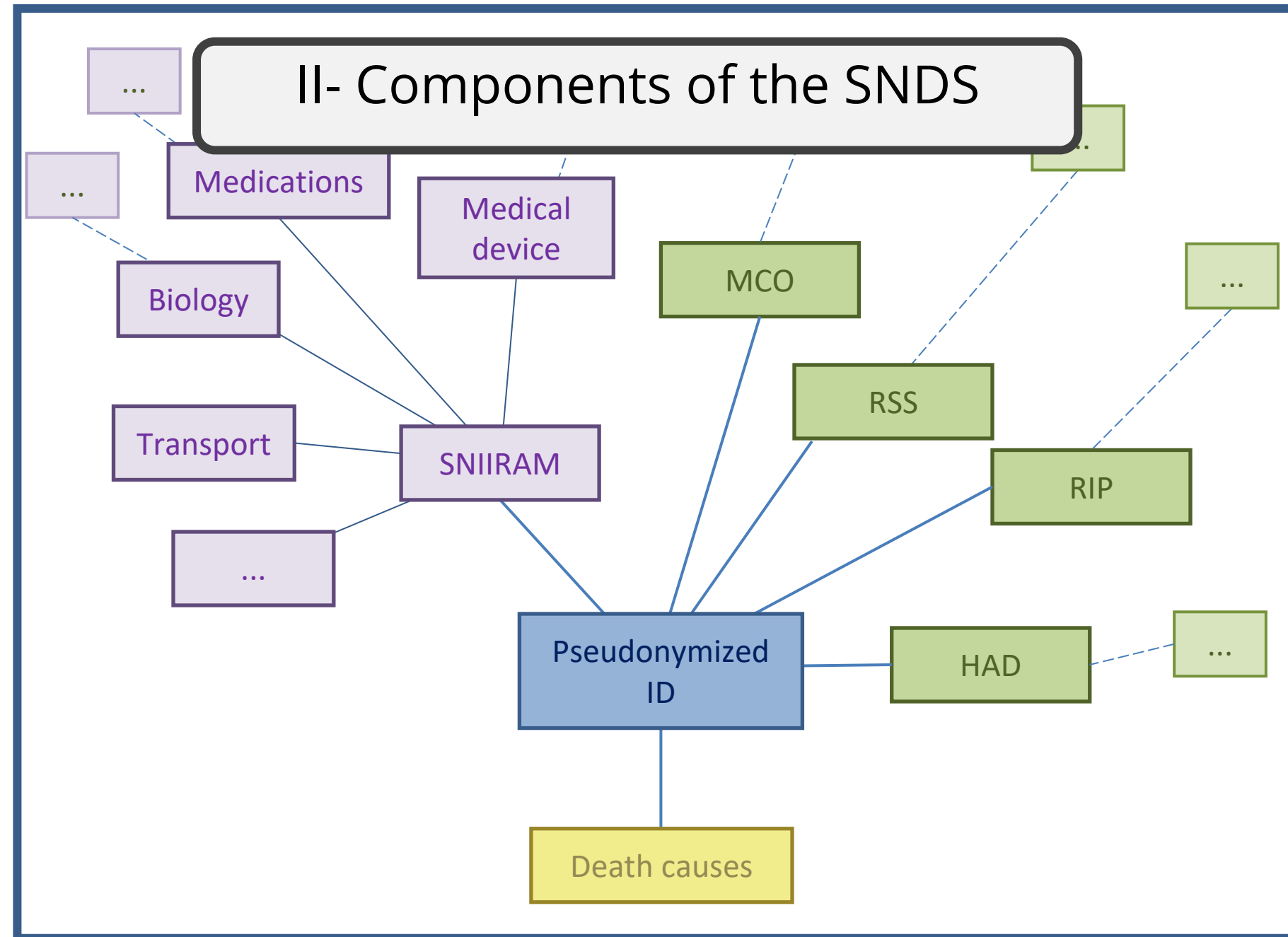
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- **Risk studies:** Do drugs imply complications to patients? What types of people and what factors are associated with the development of a disease?
- **Treatment efficacy studies**
- **Descriptive studies** of patient care

IV- Uses of the SNDS



PubMed®

+

LiSSa
Littérature Scientifique en Santé

+

HAL
science ouverte

Source:
<https://ecosysteme-snds.health-data-hub.fr>

2832

*Publication on the
SNDS in PubMed,
LiSSa & Hal*

7879

*Different
authors*

395*

*Different
structures*

*Nombre approximatif,
calculé grace aux
affiliations des auteurs

27*

Countries

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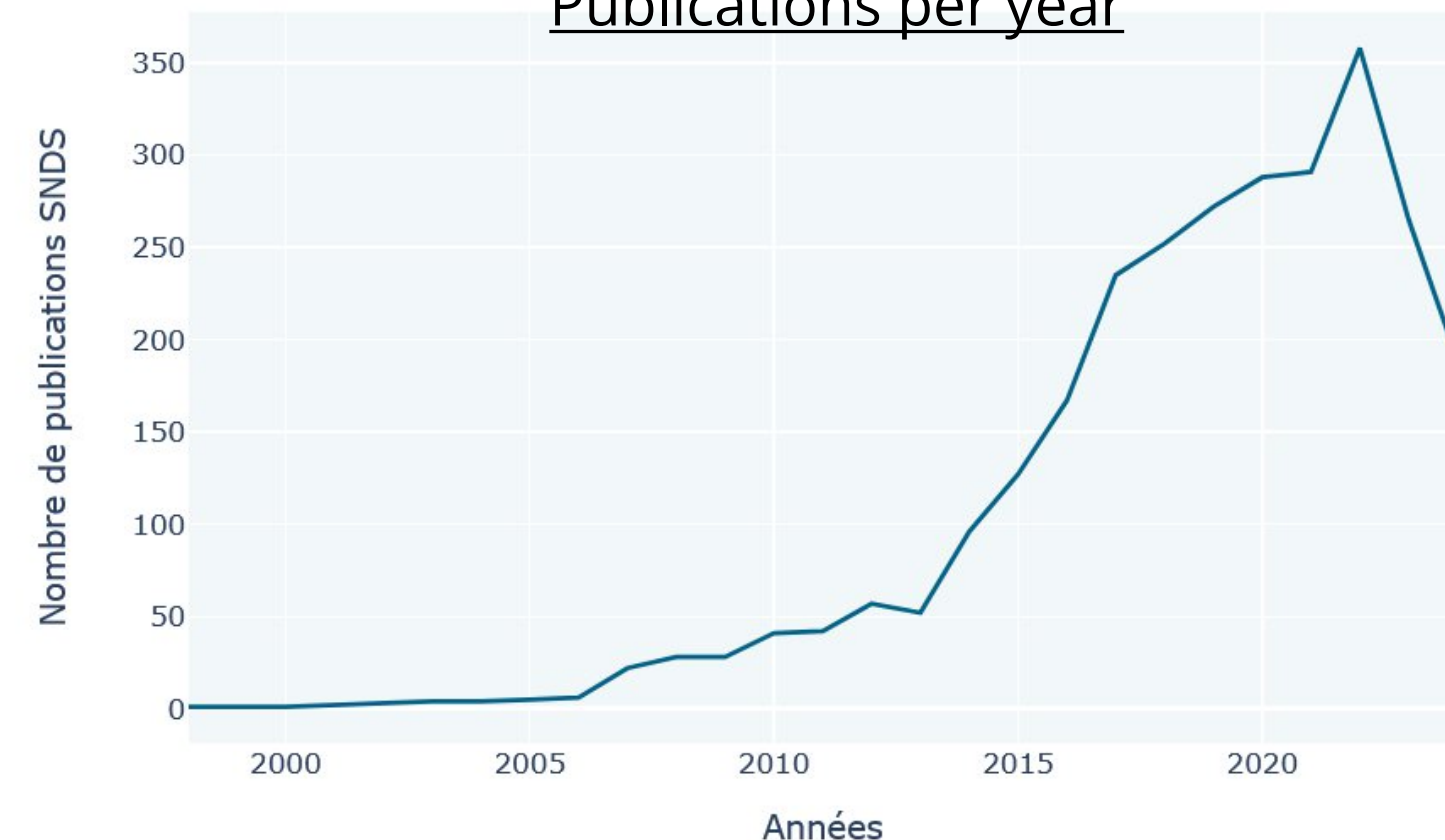
13.36

*Mean citation
per article*

A wide range of medical fields

- Cancer
- End of life
- Heart disease
- Cholestérol
- Covid-19
- Public health costs
- ...

Publications per year





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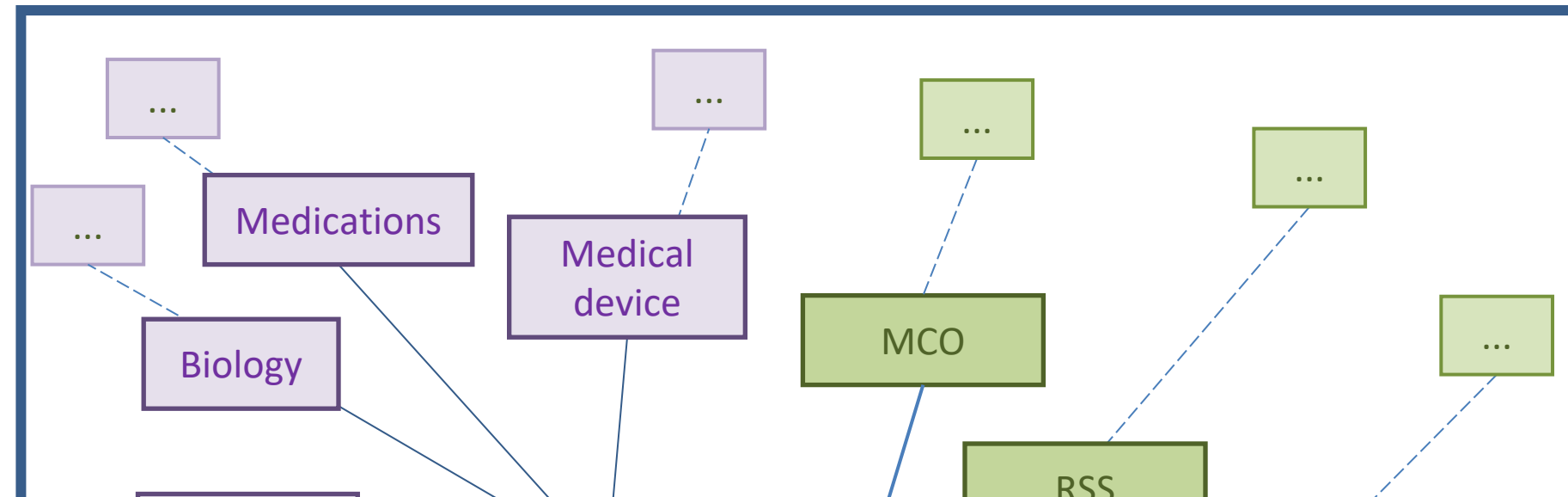
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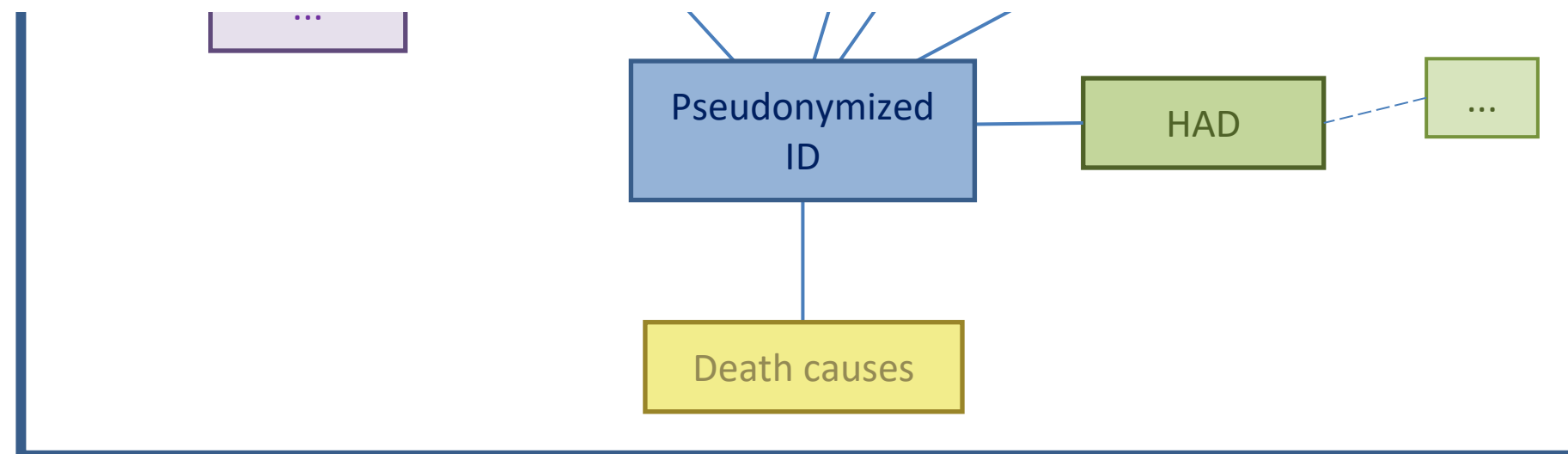
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